# TILINGS OF $W_M$ AND $T_M$ BELOW THE THRESHOLD SCALE: A SIX-PIECE TEMPLATE

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ABSTRACT. For coprime integers $2 \le e \le f-2$ let the *tile* be the triangle with sides $ef$, $f^2-e^2$, $f^2$, and let $W_M$ be a certain triangle of integer scale $M$ built from it. Paliy showed that $W_M$ is tiled by copies of the tile when $M \ge f$. We prove that $W_M$ is tiled already when $Me \ge f$ and $(M-1)(f-e) \ge f$. The proof is one layout of six pieces in a skew frame with integer coordinates, with a written proof of about four pages; a Lean 4 proof of the same statement (verifying the same layout by cross-product tables) also exists. Example: $W_4(3,5)$ is tiled by 656 tiles. For $e < f/2$ the condition is also necessary *if one assumes an exclusion theorem imported from the earlier release*. A neighbouring triangle $T_M$ is tiled by a gluing lemma of that release. A program count (computed, not proved) gives 26 138 integers, undecided in this project's census, that are tile counts. The paper states which parts are Lean, written by hand, computed or imported.

**Key (everything needed to read §1.2; every symbol is defined again in §2).** Coprime integers $2 \le e \le f-2$; $d = f-e$; tile sides $a = ef$, $b = f^2-e^2$, $c = f^2$; $\kappa = 2f^2-e^2$. A triangle of *scale* $s$ has sides $sa, sb, sc$ (tiled by $s^2$ tiles). $W_M$ is the triangle $OAB$ with $|OB| = Mfb$, $|OA| = Mf^3$, $|AB| = Me\kappa$; $P$ is the point of $AB$ with $OP = OB$. *Frame:* origin $P$, axes along $PO$ and $PB$; a point is $(\xi, y)$, a vector has squared length $\xi^2 + y^2 + \frac{e}{f}\xi y$; $O = (fbM, 0)$, $B = (0, ebM)$, $A = (0, -ef^2M)$ (all points of the layout have integer coordinates, Table 3). *Lattice lines:* with $\varphi = f\xi + ey$, the line $\ell_k$ is $\varphi = efbk$ ($k \in \mathbb{Z}$); it has direction $(e, -f)$ and meets $PO$ at $\xi = ebk$ and $AB$ at $y = fbk$. $\ell_t$ and $\ell_r$ are the two lattice lines on either side of $B$ ($ft < eM < fr$), and the *strip* $S$ is the part of $OPB$ between them; its width is one tile-row ($= af$ along $OB$) when $f \nmid eM$. Lattice coordinates $u = \varphi/efb$, $w = (e\xi + fy)/efb$ tell on which side of a line a point is. $n = \lfloor eM/f \rfloor$; canonical $t = n - \delta$, $r = n+1$, $q = n$ ($\delta = 1$ iff $f \mid eM$). (H1) $Me \ge f$, (H2) $(M-1)d \ge f$.

## 1. INTRODUCTION

1.1. **The result.** A triangle $\Omega$ is *tiled by $N$ copies of a triangle* $\mathsf{tile}$ if $\Omega$ is the union of $N$ triangles with pairwise disjoint interiors, each congruent to $\mathsf{tile}$ (mirror images allowed, no edge-to-edge condition). An integer $N$ is a *tile count* if some triangle is tiled by $N$ copies of some triangle; Erdős asked which integers are tile counts (Problem 634 of `erdosproblems.com`). Paper W constructs tilings; nothing about primes is used or claimed (the tiles with $3\alpha + 2\beta = \pi$ are the ones on which the base-$\beta$ approach of Beeson–Roşu and Paliy to the prime case rests; here each tiling of a target is only a new tile count).

For coprime integers $1 \le e < f$ put $a = ef$, $b = f^2-e^2$, $c = f^2$; the triangle $\mathsf{tile}(e,f)$ with sides $a, b, c$ has angles $\alpha, \beta, \gamma$ opposite $a, b, c$ and $3\alpha + 2\beta = \pi$. Beeson–Roşu and Paliy define two target triangles, $W_M(e,f)$ and $T_M(e,f)$, for each integer scale $M \ge 1$ (Section 2.3); their area is $M^2(2f^2-e^2)$ resp. $M^2(3f^2-e^2)$ times the area of the tile. Before this paper both were known to be tiled for every $M \ge f$ (Paliy), and for $e \ge 2$ the expectation was that no smaller $M$ works.

*Date*: Paper W, version 3.0, 19 October 2026.

**Theorem W** (Theorem 3.1). If $2 \le e \le f-2$, $\gcd(e,f)=1$, $Me \ge f$ and $(M-1)(f-e) \ge f$, then $W_M(e,f)$ is tiled by $M^2(2f^2-e^2)$ copies of $\mathsf{tile}(e,f)$ and $T_M(e,f)$ by $M^2(3f^2-e^2)$.

Example: $(e,f)=(2,5)$, $M=3$ gives a tiling of a triangle with sides $315, 276, 375$ by 414 copies of $(10,21,25)$; before, the smallest scale known was $M=f=5$. *Why it matters.* The scales $M<f$ are the ones not covered by Paliy's theorem, and the earlier release listed $W_4(3,5)$ as an open case; each tiling shows that its tile count is an integer of the kind Erdős's question asks about.

*What this does not do.* It does not settle which $M$ fail when $e > f/2$ (Section 6), the members $e=1$ and $e=f-1$, or the prime case of Problem 634.

1.2. **Reader's guide.** *The proof in outline* (Section 4; every claim below is an equation or inequality a reader can check by hand; "scale $s$" means a triangle with sides $sa, sb, sc$, and a "tile-row" is a strip one scale-one tile wide). (1) Put the origin at the point $P$ of the side $AB$ with $OP=OB$, and use as axes the lines $PO$ and $PB$ (Section 2.4); in this skew frame the three vertices of $W_M$ and all other points below have *integer* coordinates, and a vector $(\xi, y)$ has squared length $\xi^2+y^2+\frac{e}{f}\xi y$. (2) Five directions occur; for each, one integer vector and its length are listed (Table 2), so every side length is read off as "integer × length of a direction vector". (3) The line $OP$ cuts $W_M$ into the triangle $OAP$, which is a tile of the integer scale $fM$, and a triangle $OPB$. (4) The lines $\ell_k$ ($k \in \mathbb{Z}$), parallel to a side of the tile, form a lattice; $\ell_t$ and $\ell_r$ are the two lattice lines on either side of $B$, and between them lies a strip $S$ one tile-row wide. (5) Two cuts parallel to $OB$ divide $S$ into three pieces; with $\ell_t$, $\ell_r$ and the line $P_2P_3$ they give six pieces, defined by inequalities. (6) Covering is a case distinction on the signs of $y$, $u-t$, $u-r$ and $w-q$; disjointness is: two pieces lie on opposite sides of one of six lines (Section 4.3). (7) Each piece is a triangle, a trapezoid or a parallelogram of integer scale; Table 5 lists its side vectors (Section 4.4). (8) Everything above needs only the four inequalities (K): $ft < eM$, $eM < fr$, $er < fq$, $eq < ft$ for the integer parameters $t, r, q$ (Section 4.1); these four inequalities have a solution exactly when $(f-e)\lfloor eM/f \rfloor \ge e+1$, equivalently when $Me \ge f$ and $(M-1)(f-e) \ge f$ (Lemma 3.5, Proposition 4.1). (9) The tile counts of the six pieces add up to $M^2(2f^2-e^2)$ (one identity).

*How the layout was found.* A computer search (our engine, run on the $M<f$ cases left open in the earlier release) produced tilings whose pieces were always the same six shapes; the proof below says why: the bisector $OP$ makes one big tile, and what is left near $B$ is a strip that one row of lattice-line pieces fills.

*What is proved where.*

| claim | source |
|---|---|
| $W_M(e,f)$ is tiled (Thm 3.1) | Lean `thmW` (`WTilable`); and the hand proof of Section 4, a different argument |
| $M^2(2f^2-e^2)$, $M^2(3f^2-e^2)$ are tile counts | Lean `thmW` |
| $T_M(e,f)$ is tiled | glue (I5): Lean, earlier release |
| (H1)∧(H2) ⇔ $(f-e)\lfloor eM/f \rfloor \ge e+1$; the layout exists iff this holds | Lemma 3.5, Prop. 4.1: hand proofs (a few lines), exact enumeration; not Lean |
| Sharpness for $e < f/2$ | "if": this paper; "only if": imported (I1) |
| Explicit tilings (414, 656, …) | exact checker ([COMPUTED, EXACT]), not Lean |
| 26 138 new tile counts | computer count ([COMPUTED, EXACT]); each value is an instance of Theorem 3.1 |
| $W_2(5,8)$, other open scales | see Section 6; evidence only |

1.3. **Labels, credit, reading.** Every statement carries a label: [PROVED IN LEAN] a theorem of the accompanying Lean 4 library, checked by the kernel with the standard axioms only (`propext`, `Classical.choice`, `Quot.sound`), no `sorry`; the Lean statement was compared with the written one *inside this project only*; [PROVED BY HAND] a written proof in which every step is displayed; the displayed identities and inequalities of Sections 2–4 were re-checked by exact enumeration (Appendix A), no outside reader has checked them; [COMPUTED, EXACT] an exact finite computation (for an explicit tiling: exact coordinates and an exact checker); [CONJECTURE] unproved. Credits are in Section 7; the work was done with an AI system (disclosed there). *Reading.* Ten minutes: this section, Section 2, the point table (Table 3), Figure 3 and Example 5.1. An hour: Sections 2–5. Appendix A ("verification aid") is not needed to follow the proof.

## 2. The frame and the notation

2.1. **Symbols (reference table).** Table 1 lists every symbol of this paper; each is also defined where it first appears.

TABLE 1. Symbols. This table is shared word for word with the companion paper [6].

| symbol | meaning ([W] = Paper W; [C] = the companion paper) |
|---|---|
| *used in both papers* | |
| $e < f$ | coprime positive integers defining the tile; $d = f - e$ |
| $a, b, c, \kappa$ | $a = ef$, $b = f^2 - e^2$, $c = f^2$, $\kappa = b + c = 2f^2 - e^2$: the sides of the tile $\mathsf{tile}(e, f)$ and the sum $b + c$ |
| $\alpha, \beta, \gamma$ | angles of the tile opposite $a, b, c$; $3\alpha + 2\beta = \pi$, $\gamma = 2\alpha + \beta$ |
| scale $s$ | a triangle with sides $sa, sb, sc$ has *scale* $s$ and is tiled by $s^2$ tiles (ladder subdivision, [W]) |
| $W_M$, $T_M$ | the two targets of [W]; $M$ is the *scale* of the target; $W_M$ is tiled by $M^2(2f^2 - e^2)$ tiles, $T_M$ by $M^2(3f^2 - e^2)$ |
| $M^*(e, f)$ | threshold scale $\max(\lceil f/e \rceil, \lceil f/(f-e) \rceil + 1)$ of Theorem W; (H1), (H2) are $Me \geq f$ and $(M-1)(f-e) \geq f$, together equivalent to $M \geq M^*(e, f)$ |
| *used in [W] only* | |
| $n, \rho, \delta$ | $n = \lfloor eM/f \rfloor$, $\rho = eM - fn$, $\delta = 1$ if $\rho = 0$ and 0 otherwise |
| $t, r, q$ | the three free integers of the six-piece layout; canonical choice $t = n - \delta$, $r = n + 1$, $q = n$ |
| $(\xi, y)$, $(u, w)$ | skew frame with origin $P$ ($OP = OB$) and lattice coordinates $u = (f\xi + ey)/efb$, $w = (e\xi + fy)/efb$; $\ell_k = \{u = k\}$ (Section 2.3 of [W]) |
| $v, s_1, s_2, x, u_1, u_2$ | the integer scales of the pieces, $v = fM - er$, $s_1 = f(r - t)$, $s_2 = fr - eM$, $x = ft - eq$, $u_1 = fq - et$, $u_2 = fq - er$ |
| (K) | the four inequalities $ft < eM < fr$, $er < fq$, $eq < ft$ |
| $\pi_0, \ldots, \pi_5$ | the six pieces of the layout; $O, A, B, P, P_2, P_3, Q, R, B', C', V_2, X_4, U_1, U_2$ its points |
| *used in [C] only* | |
| $B_g$ | the isosceles target with base angles $\alpha$; $g$ is its scale; $b\kappa g^2$ tiles |
| row $(e, f)$ | the tile $\mathsf{tile}(e, f)$; $G(e, f)$ is the set of scales $g$ for which $B_g(e, f)$ is tiled |
| $F_0, \ldots, F_8$ | the families of Conjecture C |

2.2. **The tile and the ladder.** Throughout, $e < f$ are coprime positive integers and

$$a = ef, \qquad b = f^2 - e^2, \qquad c = f^2, \qquad \kappa = b + c = 2f^2 - e^2, \qquad d = f - e.$$

The *tile* $\mathsf{tile}(e, f)$ has sides $a, b, c$ (a triangle since $a + b - c = ed > 0$); $\alpha, \beta, \gamma$ are its angles opposite $a, b, c$. One checks $c^2 = a^2 + bc$, and $3\alpha + 2\beta = \pi$, $\gamma = 2\alpha + \beta$. The only angle used below is

$$\cos\beta = \frac{a^2 + c^2 - b^2}{2ac} = \frac{e(3f^2 - e^2)}{2f^3}.$$

Examples: $(e, f) = (2, 5)$ gives the tile $(10, 21, 25)$; $(3, 5)$ gives $(15, 16, 25)$; $(1, 2)$ gives $(2, 3, 4)$.

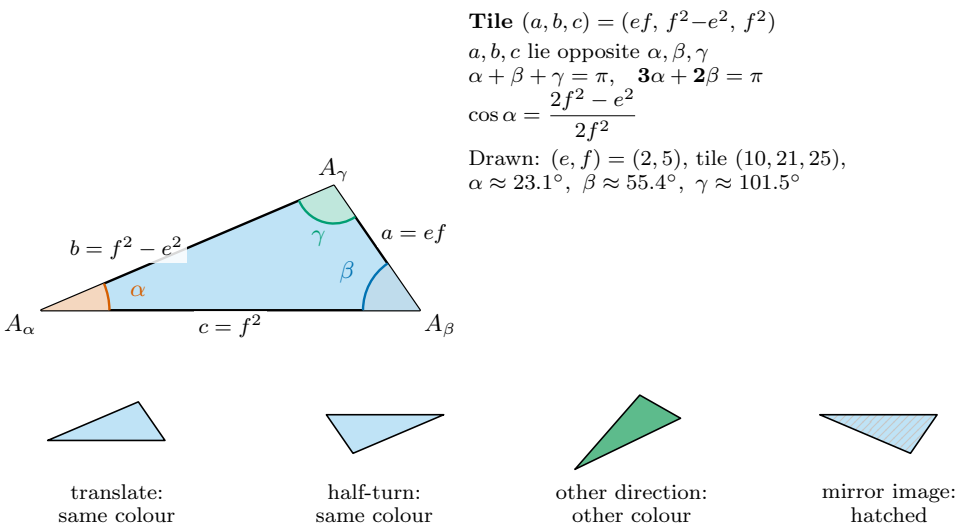



FIGURE 1. The tile $\mathsf{tile}(e, f)$ for $(e, f) = (2, 5)$, the triangle $(10, 21, 25)$: sides $a = ef, b = f^2 - e^2, c = f^2$ and angles $\alpha, \beta, \gamma = 2\alpha + \beta$ opposite them; $3\alpha + 2\beta = \pi$. In tiling figures, tiles of the same orientation (up to a half-turn) share a colour and mirror images are hatched.

A triangle with sides $sa, sb, sc$ has *scale* $s$ (a copy of the tile enlarged by $s$; by SSS it is determined by its side lengths).

**Proposition 2.1** (Ladder subdivision; [PROVED IN LEAN]). *Let $s, u \geq 1$ be integers. A triangle of scale $s$ is tiled by $s^2$ copies of* $\mathsf{tile}$. *A parallelogram with sides $sa$ and $uc$ enclosing the angle $\beta$ is tiled by $2su$ copies.*

*Lean:* `Subdivision.ladderDissection`, `Ladder.ladder` (earlier release, `thm:ladder`). *Why:* the lines parallel to the sides at integer steps cut a triangle of scale $s$ into $s^2$ triangles of scale one, all congruent to the tile (half of them rotated by $\pi$); a parallelogram with sides $a, c$ and angle $\beta$ is two tiles, since its short diagonal has length $b$ (law of cosines, $\cos\beta$ above), and a parallelogram with sides $sa, uc$ is $su$ of those. A *trapezoid* below is a triangle of scale $s_1$ minus a corner triangle of scale $s_2 < s_1$ cut off by a line parallel to the opposite side; it is tiled by $s_1^2 - s_2^2$ copies (the corner is the union of $s_2^2$ cells of the ladder).

2.3. **The targets.** For an integer $M \geq 1$ the two targets are the triangles

| target | sides | angles (for orientation) | tiles by area |
|---|---|---|---|
| $W_M(e, f)$ | $Mfb,\ Me\kappa,\ Mf^3$ | $2\alpha$ (opposite $Me\kappa$), $\beta$ (opposite $Mfb$), $\alpha+\beta$ (opposite $Mf^3$) | $M^2(2f^2 - e^2)$ |
| $T_M(e, f)$ | $Mf^3,\ Mf^3,\ Me(3f^2 - e^2)$ | isosceles; apex $3\alpha$, base angles $\beta$ | $M^2(3f^2 - e^2)$ |

(The last column is the area of the target divided by the area of the tile, so any tiling has exactly that many tiles.) $T_M$ is $W_M$ with one tile of scale $Mf$ glued on the side $Mfb$ (Section 3.3); both are drawn in Figure 2. Name the vertices of $W_M$ so that $|OB| = Mfb$, $|OA| = Mf^3$, $|AB| = Me\kappa$; the angle at $O$ is $2\alpha$.

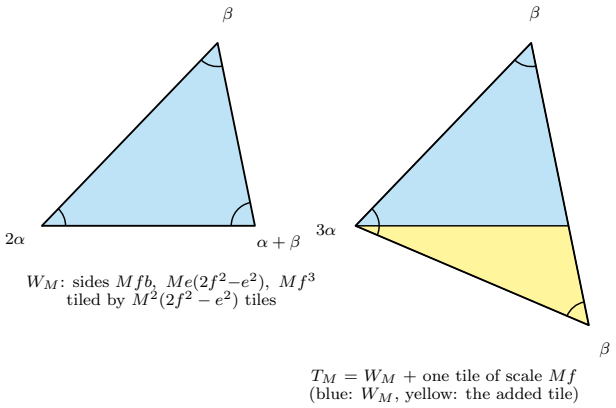



FIGURE 2. The two targets, with the proportions of $(e, f) = (2, 5)$, $M = 1$. Left: $W_M$ with angles $2\alpha$ (at $O$), $\beta$ (at $A$), $\alpha+\beta$ (at $B$). Right: $T_M$, obtained from $W_M$ by adding one tile of scale $Mf$ (yellow) along the side $Mfb$.

2.4. **The frame.** Let $P$ be the point of $AB$ with $|OP| = |OB|$. Then $|PA| = Mef^2$ and $|PB| = |AB| - |PA| = Me\kappa - Mef^2 = Meb$ (so $P$ lies between $A$ and $B$ and $PB = ebM$, $PA = ef^2M$). In the isosceles triangle $OPB$ the angle at $P$ equals the angle at $B$, and $\cos\angle OPB = \frac{|PB|/2}{|OB|} = \frac{e}{2f}$. *The frame.* Use $P$ as origin, the unit vector from $P$ towards $O$ as first axis and the unit vector from $P$ towards $B$ as second axis. A point has coordinates $(\xi, y)$, and a vector $(\xi, y)$ has (law of cosines)

$$\text{length}^2 = \xi^2 + y^2 + \tfrac{e}{f}\,\xi y. \tag{1}$$

Then $O = (fbM, 0)$, $B = (0, ebM)$, $A = (0, -ef^2M)$, and (1) gives $|OB|^2 = b^2M^2(f^2 + e^2 - e^2) = (fbM)^2$, $|OA|^2 = M^2f^2(b^2 + e^2f^2 + e^2b) = (f^3M)^2$ (because $b^2 + e^2b + e^2f^2 = f^4$), $|AB| = ef^2M + ebM = Me\kappa$: this is $W_M$. *Five directions.* Every side below is parallel to one of five lines, with primitive integer vector and length (by (1)): *Lattice coordinates.* Put $\varphi = f\xi + ey$

TABLE 2. The five directions of the layout. The length of the vector is exact; a segment in that direction with integer coordinates is an integer multiple of it.

| line | vector | length | check |
|---|---|---|---|
| $OP$, first axis | $(1, 0)$ | 1 | |
| $AB$, second axis | $(0, 1)$ | 1 | |
| $OB$ and the cuts $QV_2$, $U_1U_2$ | $(f, -e)$ | $f$ | $f^2 + e^2 - e^2 = f^2$ |
| the lines $\ell_k$ and $QU_1$, $V_2U_2$, $P_2U_1$ | $(e, -f)$ | $f$ | $e^2 + f^2 - e^2 = f^2$ |
| $OA$ and $P_2P_3$ | $(b, ef)$ | $f^2$ | $b^2 + e^2f^2 + e^2b = f^4$ |

and $\psi = e\xi + fy$, and

$$u = \frac{\varphi}{efb}, \qquad w = \frac{\psi}{efb}, \qquad \text{so that} \qquad \xi = ef(fu - ew), \quad y = ef(fw - eu).$$

For an integer $k$ the line $\ell_k$ is $\{u = k\}$; it has direction $(e, -f)$. It meets $PO$ ($y = 0$) at $\xi = ebk$ and $AB$ ($\xi = 0$) at $y = fbk$. The side $OB$ is $\{w = M\}$, the line $AB$ is $\{\xi = 0\} = \{ew = fu\}$, the line $OP$ is $\{y = 0\} = \{eu = fw\}$. $u, w$ are used only to say which side of a line a point lies on; lengths are always computed in $(\xi, y)$.

2.5. **The points of the layout.** The layout depends on the scale $M$ and on three integers $t, r, q$ (chosen in Section 4.1); the integer *scales* of its pieces are

$$v = fM - er, \quad s_1 = f(r-t), \quad s_2 = fr - eM, \quad x = ft - eq, \quad u_1 = fq - et, \quad u_2 = fq - er. \tag{2}$$

(Every entry of the $(u, w)$ columns follows from $u = \varphi/efb$, $w = \psi/efb$; for instance $u(U_1) = \big(f \cdot ef(ft - eq) + e \cdot ef(fq - et)\big)/efb = \big(ef \cdot bt\big)/efb = t$.) The earlier version of this paper described the same points by fourteen complex numbers; the dictionary is in Appendix A.3.

TABLE 3. The points, in the frame $(\xi, y)$ (integers) and in lattice coordinates $(u, w)$. $R$ and $X_4$ are apexes of the corners that are removed from two pieces, not vertices of pieces.

| point | $(\xi, y)$ | $(u, w)$ | point | $(\xi, y)$ | $(u, w)$ |
|---|---|---|---|---|---|
| $P$ | $(0, 0)$ | $(0, 0)$ | $V_2$ | $\big(ef^2(r-t),\ f(f^2t - e^2r)\big)$ | $(r,\ ft/e)$ |
| $O$ | $(fbM, 0)$ | $(fM/e,\ M)$ | $X_4$ | $(fbq, 0)$ | $(fq/e,\ q)$ |
| $B$ | $(0, ebM)$ | $(eM/f,\ M)$ | $U_1$ | $\big(ef(ft - eq),\ ef(fq - et)\big)$ | $(t,\ q)$ |
| $A$ | $(0, -ef^2M)$ | $(-\frac{efM}{b}, -\frac{f^2M}{b})$ | $U_2$ | $\big(ef(fr - eq),\ ef(fq - er)\big)$ | $(r,\ q)$ |
| $P_2$ | $(bet, 0)$ | $(t,\ et/f)$ | $C'$ | $(ber, 0)$ | $(r,\ er/f)$ |
| $P_3$ | $(0, -e^2ft)$ | $(-\frac{e^2t}{b}, -\frac{eft}{b})$ | $B'$ | $\big(ef(fr - eM),\ ef(fM - er)\big)$ | $(r,\ M)$ |
| $Q$ | $(0, fbt)$ | $(t,\ ft/e)$ | $R$ | $(0, fbr)$ | $(r,\ fr/e)$ |

## 3. Theorem W and its consequences

3.1. **Statement.**

**Theorem 3.1** (Theorem W; $W$ half [proved in Lean] and [proved by hand]; count of $T_M$ [proved in Lean]; tiling of $T_M$ via the imported glue)**.** *Let $e, f$ be coprime integers with $2 \le e \le f - 2$ and let $M \ge 1$ be an integer with*

$$\textit{(H1)} \quad Me \ge f, \qquad\qquad \textit{(H2)} \quad (M-1)(f-e) \ge f.$$

*Then $W_M(e, f)$ is tiled by $M^2(2f^2 - e^2)$ copies of* tile$(e, f)$*, and $T_M(e, f)$ is tiled by $M^2(3f^2 - e^2)$ copies.*

Put $M^*(e, f) = \max\big(\lceil f/e \rceil,\ \lceil f/(f-e) \rceil + 1\big)$; then (H1) and (H2) together say $M \ge M^*(e, f)$.

*Exactly what is Lean.* Lean module `Draft/ThmW.lean` (with 19 helper modules, aggregate `Erdos634.AllThmW`), theorem `Draft.ThmW.thmW`: for natural numbers $e, f, M$ with `Nat.Coprime e f`, $2 \le e$, $e + 2 \le f$, $f \le M \cdot e$ and $f \le (M-1) \cdot (f-e)$, the conclusion is

```
WTilable e f M ∧ TileCount ((2*f^2-e^2)*M^2) ∧ TileCount (M^2*(3*f^2-e^2)).
```

`WTilable e f M` says: there is a dissection into congruent triangles, each congruent to the triangle with sides $ef,\ f^2 - e^2,\ f^2$, of a triangle with sides $Mf(f^2 - e^2)$, $Me(2f^2 - e^2)$, $Mf^3$. So Lean states the *tiling* for $W_M$; for $T_M$ it states only that the number is a tile count, and the tiling of $T_M$ itself is the glue (I5), a Lean theorem of the earlier release (Section 3.3). Axioms: `propext`, `Classical.choice`, `Quot.sound`. The Lean statement was compared with the one above inside the project (verdict: match). Hypotheses are witnessed by instances `inst_414`, `inst_656`,

`inst_801` (and others), so the Lean theorem is not vacuous. *The Lean proof does not follow Section 4*: it verifies, for the layout in complex coordinates, a table of about 100 cross products factored into positive factors, with 143 generated sign lemmas (Appendix A.3). Section 4 is a second proof of the $W$ half (the same layout, a different verification of it), written for a reader; it is not formalised.

First instances (the last column is the smallest scale known before, namely $f$):

| $(e, f)$ | $M^*$ | first tiled $W_{M^*}$ | number of tiles | smallest $M$ known before |
|---|---|---|---|---|
| $(2, 5)$ | 3 | $W_3(2, 5)$ | 414 | 5 |
| $(3, 5)$ | 4 | $W_4(3, 5)$ | 656 | 5 |
| $(3, 7)$ | 3 | $W_3(3, 7)$ | 801 | 7 |
| $(2, 11)$ | 6 | $W_6(2, 11)$ | 8568 | 11 |
| $(9, 11)$ | 7 | $W_7(9, 11)$ | 7889 | 11 |

Thirteen targets with $M < f$ were found by a bounded computer search before the proof was known (Appendix B); they are only the output of that one search (bounded by the search's own parameter range, which did not reach, for example, $W_3(2, 5)$). The theorem has many more instances with $M < f$, for example $(e, f, M) = (4, 9, 3)$ and $(3, 8, 3)$.

3.2. **Consequences.**

**Corollary 3.2** (Sharpness for $e < f/2$, assuming the import (I1); "if": Theorem 3.1 [PROVED IN LEAN] plus the arithmetic below; "only if": imported (I1))**.** *Let* $\gcd(e, f) = 1$ *and* $2 \le e < f/2$. *Then* $W_M(e, f)$ *is tiled if and only if* $Me \ge f$.

*Proof.* "Only if" is (I1) (Section 3.4). "If": $e < f/2$ gives $f - e > f/2$, so $f/(f - e) < 2$ and $\lceil f/(f - e) \rceil + 1 = 3$, while $f/e > 2$ gives $\lceil f/e \rceil \ge 3$; so $M^* = \lceil f/e \rceil$. Moreover (H1) implies (H2): (H1) gives $M \ge 3$, so $(M - 1)(f - e) \ge 2(f - e) > f$. Apply Theorem 3.1. □

For $e < f/2$ the threshold is therefore *exactly* $\lceil f/e \rceil$, smaller than $f$ as soon as $e \ge 2$. (For $e = 1$ it is $f$, by (I3).) The "only if" half is not proved in this paper; it is only as strong as the import (I1).

**Corollary 3.3** (An open case; [PROVED IN LEAN] for the tiling, [COMPUTED, EXACT] for the data)**.** $W_4(3, 5)$*, a triangle with sides* 320, 492, 500*, is tiled by* 656 *copies of* $(15, 16, 25)$.

This is Theorem 3.1 for $(e, f, M) = (3, 5, 4)$ ($M^*(3, 5) = \max(2, 4) = 4$). A list of the 656 tiles in exact coordinates is shipped (Appendix B) and verified by an exact checker.

**Remark 3.4** (An earlier expectation that is false)**.** (a) *The threshold conjecture "B1".* Our earlier notes expected that, for $e \ge 2$, neither $W_M$ nor $T_M$ is tiled for $M < f$; the earlier paper [4] recorded as an observation that every case decided so far has threshold $f$, and listed $W_4(3, 5)$ among the open cases. Corollary 3.3 is a counterexample ($4 < 5$), and Theorem 3.1 gives infinitely many (for each $e \ge 2$ and $f$ large, $M = \lceil f/e \rceil < f$ works when $e < f/2$, by Corollary 3.2). The first statement was an expectation of our working notes and an observation in the earlier paper, not a theorem; no earlier theorem is affected.

3.3. **The $T$ half.** Glue to the side $Mfb$ of $W_M$ (the side $OB$) one copy of the tile at scale $Mf$, with its side $Mfb$ on that side and its third vertex $Z$ on the line $AB$ beyond $B$, so that $BZ = Mef^2$. The result is the isosceles triangle with legs $Mf^3$ and base $Me(3f^2 - e^2)$, that is $T_M$, and its tile count is $M^2(2f^2 - e^2) + (Mf)^2 = M^2(3f^2 - e^2)$. This is (I5). The Lean theorem `Draft.ThmW.thmW` contains the *count* for $T_M$, obtained from the tiling of $W_M$ by `DebtB.counts_of_wTilable`, which needs besides the tiling only $M(f - e) \ge f$; (H2) implies it. The converse (a tiling of $T_M$ gives one of $W_M$) is not known, and the proved exclusions for

$T_M$ are weaker ((I1): only $2Me < f$). Hence the label of the $T$ half: tiling of $W_M$ [PROVED IN LEAN] [PROVED BY HAND]; tile count of $T_M$ [PROVED IN LEAN]; tiling of $T_M$ by the imported glue [PROVED IN LEAN] in the earlier release.

3.4. **Five facts imported from the earlier release.** Five facts about the tile and the targets are Lean theorems of the earlier release [4, 5] and are used as stated, not reproved. *Tiled* means tiled by copies of $\mathsf{tile}(e, f)$ with the same $(e, f)$.

- (I1) (`cor:ebelow`(a), (b), and the case $e = 1$) If $Me < f$ then $W_M(e, f)$ is not tiled; if $2Me < f$ then $T_M(e, f)$ is not tiled; for $e = 1$, $T_M(1, f)$ is not tiled if $2M \le f$. [All coprime $1 \le e < f$.]
- (I2) (`cor:thickw`(a)) If $M(f - e) \le e$ then $W_M(e, f)$ is not tiled.
- (I3) (`cor:below`, `cor:thickw`(b)) For $e = 1$ and for $e = f - 1$: $W_M(e, f)$ is tiled if and only if $M \ge f$.
- (I4) (Paliy; formalised there) For every $M \ge f$, $W_M(e, f)$ and $T_M(e, f)$ are tiled.
- (I5) (the glue; Lean `CollarG.betaTilable_of_wTilable` and `DebtB.counts_of_wTilable`) If $W_M(e, f)$ is tiled then $T_M(e, f)$ is tiled; and if $W_{M_0}(e, f)$ is tiled and $M_0(f - e) \ge f$, then for every $M \ge M_0$ both $W_M$ and $T_M$ are tiled and $M^2(2f^2 - e^2)$, $M^2(3f^2 - e^2)$ are tile counts.

They are used only outside the proof of the $W$ half: (I1) for Corollary 3.2 and to say what is open; (I2), (I3) to say what is open and in Remark 3.4; (I4) to place the theorem against what was known; (I5) for the $T$ half. The proof of the $W$ half (Section 4) uses none of them.

3.5. **An equivalent form of the hypotheses.** Write $n = \lfloor eM/f \rfloor$, $\rho = eM - fn \in [0, f)$ and $\delta = 1$ if $\rho = 0$, $\delta = 0$ otherwise.

**Lemma 3.5** ([PROVED BY HAND]; not in Lean)**.** *Let $2 \le e \le f - 2$, $\gcd(e, f) = 1$, $d = f - e$, $M \ge 1$. Then (H1) and (H2) hold if and only if $dn \ge e + 1$.*

*Proof.* First, $ed \ge f+1$: with $f = e+d$ we have $ed-f = (e-1)(d-1)-1$, and $(e-1)(d-1) \ge 2$ because $e, d \ge 2$ and $e = d = 2$ would give $f = 4$, not coprime to $e = 2$. $(*)$

($\Leftarrow$) Suppose $dn \ge e+1$. Then $n \ge 1$, i.e. $eM \ge f$: (H1). Suppose (H2) failed: $(M-1)d \le f-1$. Then $dM \le f+d-1 < 2f$, so $s := \lceil dM/f \rceil \in \{1, 2\}$, and $n = \lfloor M - dM/f \rfloor = M - s$. If $s = 1$ then $dM \le f$ and $dn = d(M-1) \le f-d = e$; if $s = 2$ then $dn = d(M-2) \le f+d-1-2d = e-1$. Either way $dn \le e$, a contradiction.

($\Rightarrow$) Suppose (H1), (H2). If $d \ge e + 1$: $n \ge 1$ by (H1), so $dn \ge d \ge e + 1$. If $d \le e$: $d \nmid f$ (else $d \mid e$, so $d = 1$, i.e. $e = f - 1$, excluded), so $f = dh + j$ with $0 < j < d$, and $h \ge 2$ because $f = e + d \ge 2d$; then $e = f - d = d(h-1) + j$. (H2) gives $M - 1 \ge \lceil f/d \rceil = h + 1$, so $M \ge h + 2$, and

$$e(h + 2) - hf = e(h + 2) - h(e + d) = 2e - hd = d(h - 2) + 2j > 0,$$

so $eM \ge e(h + 2) > hf$, hence $n \ge h$ and $dn \ge dh = f - j \ge f - d + 1 = e + 1$. □

**Corollary 3.6** (Theorem W, restated; [PROVED BY HAND] (Lemma 3.5) on top of [PROVED IN LEAN])**.** *For coprime $2 \le e \le f-2$ and $M \ge 1$ with $(f-e)\lfloor eM/f \rfloor \ge e+1$, the target $W_M(e, f)$ is tiled by $M^2(2f^2-e^2)$ tiles and $T_M(e, f)$ by $M^2(3f^2-e^2)$ tiles. The tile counts of the six pieces of Section 4 are, for $\delta = 0$,*

$$f^2M^2 - e^2n^2, \quad (fM - en - e)^2, \quad f^2n^2, \quad \rho(2f - \rho), \quad 2f\,dn, \quad e(2dn - e).$$

The hypotheses of the Lean theorem stay (H1), (H2); the lemma is proved here, by hand only. Consequently $M^*(e, f)$ is the least $M$ with $d\lfloor eM/f \rfloor \ge e + 1$.

## 4. Proof of the $W$ half of Theorem W

Standing assumptions: $e, f$ coprime, $2 \le e \le f-2$, $d = f-e \ge 2$, $M \ge 1$; $t, r, q$ integers.

4.1. **The four inequalities.** We use the following condition on $(t, r, q)$:

$$\text{(K)} \qquad ft < eM < fr, \qquad er < fq, \qquad eq < ft. \tag{3}$$

In lattice coordinates, (K) says: on the line $AB$ the point $B$ lies between the lattice lines $\ell_t$ and $\ell_r$, and on the lines $\ell_t, \ell_r$ the level $q$ lies in the window $er/f < q < ft/e$. Consequences used below (each a one-line derivation):

$$\begin{gathered} t < r, \quad t \ge 1, \quad q \ge 1, \quad q < M, \\ e^2 r < f^2 t, \quad er < fM, \quad et < fq, \quad et < fM. \end{gathered} \tag{4}$$

(Derivations: $ft < eM < fr$ gives $t < r$; $er > 0$ and $er < fq$ give $q \ge 1$; $eq < ft < eM$ gives $q < M$; $eq > 0$ gives $t \ge 1$; $e^2 r < efq < f^2 t$; $er < fq < fM$; $et < er < fq$; $et < eM < fM$.) All six scales (2) are positive and $s_1 > s_2$, $u_1 > u_2$: $v = fM - er > 0$; $s_1 = f(r-t) > 0$; $s_2 = fr - eM > 0$; $s_1 - s_2 = eM - ft > 0$; $x = ft - eq > 0$; $u_2 = fq - er > 0$; $u_1 - u_2 = e(r-t) > 0$.

**Proposition 4.1** (the layout exists exactly when $dn \ge e+1$; [proved by hand], [computed, exact] for the enumeration)**.** *For coprime $2 \le e \le f-2$ and $M \ge 1$ the following are equivalent: (i) there are integers $t, r, q$ with (K); (ii) $dn \ge e+1$ where $n = \lfloor eM/f \rfloor$; (iii) (H1) and (H2). If they hold, the triple $t = n - \delta$, $r = n+1$, $q = n$ satisfies (K). Every triple satisfying (K) has $t \le n - \delta$ and $r \ge n+1$.*

*Proof.* (ii)⇔(iii) is Lemma 3.5. (ii)⇒(i) for the stated triple: $n \ge 1$ by (ii). $ft < eM$: if $\delta = 0$ then $\rho > 0$ and $fn < fn + \rho = eM$; if $\delta = 1$ then $f(n-1) < fn = eM$. $eM < fr$: $eM = fn + \rho < fn + f$. $er < fq$: $e(n+1) < fn \Leftrightarrow dn > e$, which is (ii). $eq < ft$: if $\delta = 0$, $en < fn$; if $\delta = 1$ then $f \mid eM$, so $f \mid M$ (coprimality), $M = fm'$, $n = em'$ and $dn \ge de \ge f+1$ by $(*)$, which is $en < f(n-1)$. (i)⇒(ii) and the last sentence: from $ft < eM$ we get $t < eM/f < n+1$, so $t \le n$, and $t \le n-1$ if $\rho = 0$; from $eM < fr$ we get $r > n$, so $r \ge n+1$. Suppose $dn \le e$, that is $fn \le e(n+1)$. Then $fq > er \ge e(n+1) \ge fn$, so $q \ge n+1$, and $eq \ge e(n+1) \ge fn \ge ft > eq$, a contradiction. □

So the layout below exists for *no* integers $t, r, q$ when $dn \le e$; this is a statement about this layout, not about tilings (Section 6). [computed, exact]: for $f \le 12$ and all $M < 3f$, all integer triples with $0 \le t < r \le fM/e + 2$, $0 \le q \le M+2$ were enumerated; (K) (and the sixteen named factors of the older proof, Appendix A.3) hold for the same triples; no discrepancy.

**Remark 4.2** (why these values of $t, r, q$)**.** Only $t \le n - \delta$ and $r \ge n+1$ are forced (by $ft < eM < fr$). The admissible triples are not unique: for $W_3(2,5)$ the triples $(t, r, q) = (1,2,1), (1,2,2), (1,3,2), (1,4,2)$ all satisfy (K). The canonical triple $(n - \delta, n+1, n)$ is the extremal admissible one: it has the narrowest strip, and it is admissible whenever any triple is (Proposition 4.1). Nothing else is claimed.

4.2. **The six pieces.** Fix $(t, r, q)$ with (K). The six pieces are the following regions, written with the lattice coordinates $(u, w)$ and $(\xi, y)$ of Section 2.4; $O, A, B, \ldots$ are the points of Table 3. The line $\{ef\xi - by = \text{const}\}$ is parallel to $OA$ (direction $(b, ef)$); the one with constant $e^2 fbt$ passes through $P_2$ and $P_3$. Figure 3 shows the six pieces in the true shape of the frame for $W_3(2,5)$, with the integer lengths of the edges. Three of the cutting lines are $\ell_t$, $\ell_r$ and the horizontal ($w$-constant, direction $(f, -e)$) lines $w = ft/e$ (through $Q$) and $w = q$ (through $U_1$, $X_4$).

TABLE 4. The six pieces as intersections of half-planes; vertices; and the shape (Section 4.4).

| piece | region | vertices | shape and tile count |
|---|---|---|---|
| $\pi_0$ | $y \le 0,\ \xi \ge 0,\ e^2fbt \le ef\xi - by \le ef^2bM$ | $A, O, P_2, P_3$ | trapezoid: $OAP$ (scale $fM$) minus corner $PP_2P_3$ (scale $et$); $f^2M^2 - e^2t^2$ |
| $\pi_1$ | $y \ge 0,\ \xi \ge 0,\ u \ge r,\ w \le M$ | $O, B', C'$ | triangle, scale $v$; $v^2$ |
| $\pi_2$ | $\xi \ge 0,\ u \le t,\ ef\xi - by \le e^2fbt$ | $P_3, P_2, Q$ | triangle, scale $ft$; $f^2t^2$ |
| $\pi_3$ | $\xi \ge 0,\ t \le u \le r,\ ft/e \le w \le M$ | $B, Q, V_2, B'$ | trapezoid: $RQV_2$ (scale $s_1$) minus corner $RBB'$ (scale $s_2$); $s_1^2 - s_2^2$ |
| $\pi_4$ | $t \le u \le r,\ q \le w \le ft/e$ | $Q, U_1, U_2, V_2$ | parallelogram $as_1 \times cx$, angle $\beta$; $2s_1x$ |
| $\pi_5$ | $y \ge 0,\ t \le u \le r,\ w \le q$ | $U_1, P_2, C', U_2$ | trapezoid: $X_4U_1P_2$ (scale $u_1$) minus corner $X_4C'U_2$ (scale $u_2$); $u_1^2 - u_2^2$ |

4.3. **Step 1: the pieces cover $W_M$ and overlap only in boundaries.** Write $\Omega = OAB = \{\xi \ge 0,\ w \le M,\ ef\xi - by \le ef^2bM\}$ (the three sides $AB$, $OB$, $OA$).

*Vertices.* Each piece is the polygon whose vertices are listed in Table 4; that the listed points are the corners of the half-plane region is read off the lattice coordinates of Table 3. For $\pi_4$: the region is the rectangle $[t, r] \times [q, ft/e]$ in the $(u, w)$-plane, and $U_1 = (t, q)$, $U_2 = (r, q)$, $V_2 = (r, ft/e)$, $Q = (t, ft/e)$; it is non-degenerate since $q < ft/e$ by (K). For $\pi_5$: $P_2 = (t, et/f)$, $U_1 = (t, q)$, $U_2 = (r, q)$, $C' = (r, er/f)$, a trapezoid with two sides on $u = t$, $u = r$, since $et/f < er/f < q$ by (4). For $\pi_3$: it lies between the lines $w = ft/e$ and $w = M$ (note $ft/e < M$) and between the line $AB$ (through $Q = (t, ft/e)$, $B = (eM/f, M)$) and the line $u = r$ ($V_2 = (r, ft/e)$, $B' = (r, M)$), and $eM/f < r$. For $\pi_1$: the triangle with vertices $O = (fM/e, M)$, $B' = (r, M)$, $C' = (r, er/f)$, non-degenerate since $eM/f < r < fM/e$ (the right inequality is $er < fM$). For $\pi_2$ and $\pi_0$: the corner $P, P_2, P_3$ and the lines $P_2P_3 \parallel OA$, $AB$, $\ell_t$.

*Containment.* $\Omega$ is convex, so it suffices that every vertex lies on a segment joining points of $\Omega$:

$$P_2, C' \in [P, O], \quad P_3 \in [A, P], \quad Q \in [P, B], \quad B' \in [B, O], \quad V_2, U_2 \in [C', B'], \quad U_1 \in [P_2, Q].$$

In coordinates these are, in this order: $0 \le t, r$ and $bet, ber \le fbM$ (as $et, er < fM$); $0 \le e^2ft \le ef^2M$ (as $et < fM$); $0 \le fbt \le ebM$ (as $ft < eM$); $eM/f \le r \le fM/e$ on the line $w = M$ (as $eM < fr$, $er < fM$); on the line $u = r$ the levels $w(V_2) = ft/e$ and $w(U_2) = q$ lie in $[er/f, M]$ (as $e^2r < f^2t$, $ft < eM$, $er < fq$, $q < M$); on $u = t$ the level $w(U_1) = q$ lies in $[et/f, ft/e]$ (as $et < fq$, $eq < ft$). All are (3), (4).

*Covering.* Let $X \in \Omega$.

- If $y \le 0$: $X$ lies in $\pi_0$ if $ef\xi - by \ge e^2fbt$. Otherwise $ef\xi \le e^2fbt + by \le e^2fbt$, so $\xi \le ebt$ and $f\xi + ey \le efbt$, i.e. $u \le t$, and $X \in \pi_2$.
- If $y \ge 0$ and $u \le t$: $f\xi = efbu - ey \le efbt$, so $ef\xi - by \le e^2fbt$, and $X \in \pi_2$.
- If $u \ge r$ (and $y \ge 0$, $w \le M$): $X \in \pi_1$.
- If $y \ge 0$ and $t < u < r$: according as $w \ge ft/e$, $q \le w \le ft/e$, $w \le q$, the point lies in $\pi_3$, $\pi_4$, $\pi_5$.

So $\Omega \subseteq \bigcup \pi_i$, and by containment $\Omega = \bigcup \pi_i$.

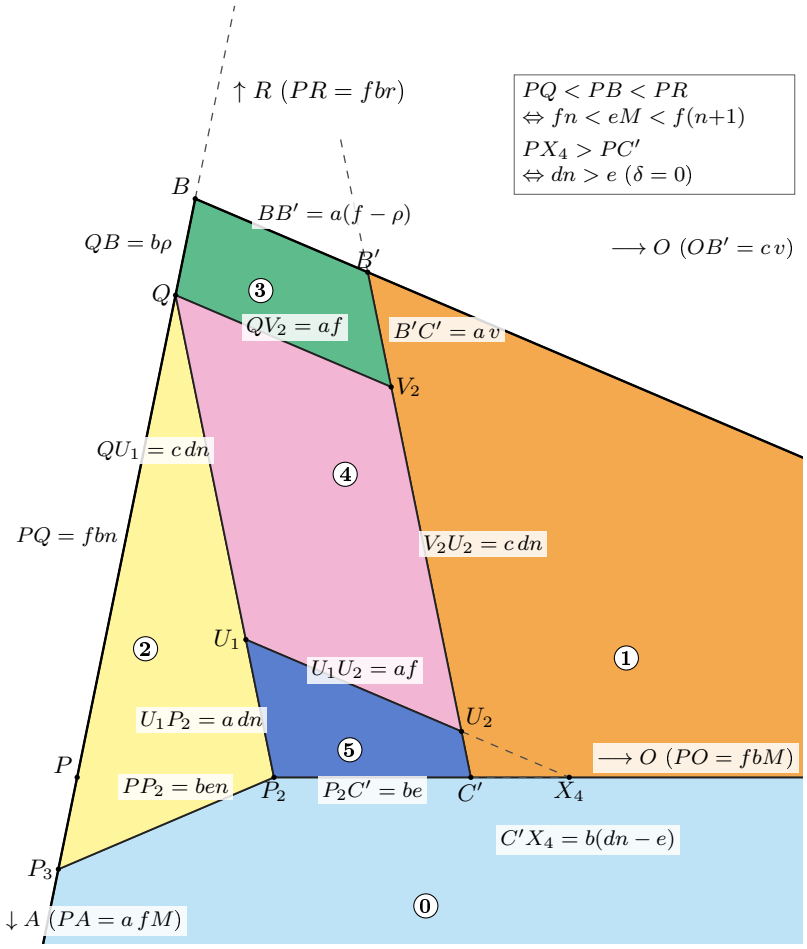



FIGURE 3. The six pieces of $W_3(2,5)$ in the true shape of the skew frame (angle at $P$ is $\arccos\frac{1}{5}$), enlarged near $P$; $\delta = 0$, $(t,r,q) = (1,2,1)$, $n = 1$, $\rho = 1$. Pieces are numbered 0–5 as in Table 4. Edge labels are exact lengths: $a = 10$, $b = 21$, $c = 25$, $d = 3$, $f = 5$, $e = 2$, $M = 3$. The inequalities in the box are the two things the layout needs: the point $B$ lies between $Q$ and $R$ (this fixes $t$ and $r$), and the apex $X_4$ lies strictly beyond $C'$ on $PO$ (this makes $\pi_5$ non-empty; $PX_4 - PC' = b(dn - e)$). $O$, $A$, $R$ lie outside the drawing; $PO = fbM$, $PA = a\,fM$.

*Disjointness, pair by pair.* For each pair the separating line, and the reason each piece lies on its side:

| pairs | line | reason |
|---|---|---|
| $\pi_0 \vert \pi_1, \pi_3, \pi_4, \pi_5$ | $y = 0$ | $\pi_0$: $y \le 0$ by definition; $\pi_1, \pi_5$: $y \ge 0$ by definition; $\pi_3, \pi_4$: $w \ge eu/f$ since $w \ge q \ge er/f \ge eu/f$ (or $w \ge ft/e \ge er/f$, using $e^2 r < f^2 t$) |
| $\pi_0 \vert \pi_2$ | $ef\xi - by = e^2 fbt$ | the definitions |
| $\pi_2 \vert \pi_1, \pi_3, \pi_4, \pi_5$ | $u = t$ | $\pi_2$: $u \le t$; $\pi_1$ ($u \ge r$), $\pi_4, \pi_5$ ($u \ge t$); $\pi_3$: $u \ge ew/f \ge t$ |
| $\pi_1 \vert \pi_3, \pi_4, \pi_5$ | $u = r$ | $\pi_1$: $u \ge r$; the others $u \le r$ |
| $\pi_3 \vert \pi_4, \pi_5$ | $w = ft/e$ | $\pi_3$: $w \ge ft/e$; $\pi_4, \pi_5$: $w \le ft/e$ (as $q \le ft/e$) |
| $\pi_4 \vert \pi_5$ | $w = q$ | $\pi_4$: $w \ge q$; $\pi_5$: $w \le q$ |

4.4. **Step 2: each piece is a union of tiles.** Table 5 lists the side vectors of every triangle used, as an integer times a primitive vector of Table 2; each line is a subtraction of two rows of Table 3. By (1) the length of a side is that integer times the length of the primitive vector, so the triangle has the stated scale (SSS). *Consequences.* (a) $OAP$ is a tile of scale $fM$ and $PP_2P_3$

TABLE 5. Side vectors. In a triangle of scale $s$ the three sides have lengths $sa$, $sb$, $sc$ (listed in the order of the vectors).

| triangle (scale) | side vectors | lengths |
|---|---|---|
| $OAP$ $(fM)$ | $O - P = fbM\,(1,0)$; $P - A = ef^2 M\,(0,1)$; $A - O = -fM\,(b, ef)$ | $bfM$; $afM$; $cfM$ |
| $PP_2P_3$ $(et)$ | $P_2 - P = bet\,(1,0)$; $P - P_3 = e^2 ft\,(0,1)$; $P_3 - P_2 = -et\,(b, ef)$ | $b{\cdot}et$; $a{\cdot}et$; $c{\cdot}et$ |
| $P_3P_2Q$ $(ft)$ | $Q - P_3 = f^3 t\,(0,1)$; $Q - P_2 = -bt\,(e, -f)$; $P_2 - P_3 = et\,(b, ef)$ | $c{\cdot}ft$; $b{\cdot}ft$; $a{\cdot}ft$ |
| $OB'C'$ $(v)$ | $C' - O = -bv\,(1,0)$; $B' - O = -fv\,(f, -e)$; $B' - C' = -ev\,(e, -f)$ | $b{\cdot}v$; $c{\cdot}v$; $a{\cdot}v$ |
| $RQV_2$ $(s_1)$ | $R - Q = fb(r-t)\,(0,1)$; $V_2 - Q = es_1\,(f, -e)$; $V_2 - R = fs_1\,(e, -f)$ | $b{\cdot}s_1$; $a{\cdot}s_1$; $c{\cdot}s_1$ |
| $RBB'$ $(s_2)$ | $R - B = bs_2\,(0,1)$; $B' - R = fs_2\,(e, -f)$; $B' - B = es_2\,(f, -e)$ (parallel to $OB$) | $b{\cdot}s_2$; $c{\cdot}s_2$; $a{\cdot}s_2$ |
| $X_4U_1P_2$ $(u_1)$ | $P_2 - X_4 = -bu_1\,(1,0)$; $U_1 - X_4 = -fu_1\,(f, -e)$; $P_2 - U_1 = eu_1\,(e, -f)$ | $b{\cdot}u_1$; $c{\cdot}u_1$; $a{\cdot}u_1$ |
| $X_4C'U_2$ $(u_2)$ | $C' - X_4 = -bu_2\,(1,0)$; $U_2 - X_4 = -fu_2\,(f, -e)$; $U_2 - C' = -eu_2\,(e, -f)$ | $b{\cdot}u_2$; $c{\cdot}u_2$; $a{\cdot}u_2$ |

is a corner of it cut off by the line $P_2P_3 \parallel OA$ at the integer step $et$: $\pi_0$ is a union of $f^2M^2 - e^2t^2$ tiles (Prop. 2.1). (b) $\pi_1$ and $\pi_2$ are triangles of scales $v$ and $ft$. (c) $RQV_2$ has scale $s_1$ and $RBB'$ is its corner at $R$ of scale $s_2$ ($B$ on $RQ$, $B'$ on $RV_2$, $BB' \parallel QV_2$, both $\parallel OB$): $\pi_3$ is a union of $s_1^2 - s_2^2$ tiles. (d) $X_4U_1P_2$ has scale $u_1$ and $X_4C'U_2$ is its corner of scale $u_2$ ($C'U_2 \parallel P_2U_1$): $\pi_5$ is a union of $u_1^2 - u_2^2$ tiles. (e) $\pi_4$ has the sides $U_1 - Q = fx\,(e, -f)$ and $V_2 - Q = es_1\,(f, -e)$, of lengths $cx$ and $as_1$, and $U_2 - V_2 = U_1 - Q$, $U_2 - U_1 = V_2 - Q$, so it is a parallelogram; the angle at $Q$ between $(f, -e)$ and $(e, -f)$ has cosine

$$\frac{\langle (f,-e),(e,-f)\rangle}{f\cdot f} = \frac{2ef + \frac{e}{2f}\left(-f^2 - e^2\right)}{f^2} = \frac{e(3f^2 - e^2)}{2f^3} = \cos\beta,$$

where $\langle (\xi, y), (\xi', y')\rangle = \xi\xi' + yy' + \frac{e}{2f}(\xi y' + y\xi')$ is the bilinear form of (1); so the angle is $\beta$ and, by Proposition 2.1, $\pi_4$ is a union of $2s_1x$ tiles.

4.5. **Step 3: the count, and the end of the proof.** By Steps 1 and 2, $W_M$ is the union of six pieces with disjoint interiors, each a union of tiles with the counts of Table 4; so $W_M$ is tiled by their sum. That sum is $M^2\kappa$: the identity

$$(f^2M^2 - e^2t^2) + v^2 + f^2t^2 + (s_1^2 - s_2^2) + 2s_1x + (u_1^2 - u_2^2) = M^2(2f^2 - e^2)$$

is a polynomial identity in $e, f, M, t, r, q$ (expand with (2); checked by computer algebra, Appendix A). It is also forced by area: the tiles are disjoint, cover $W_M$, and all have the area of the tile. For $\delta = 0$ and the canonical triple it reads, with $eM = fn + \rho$, $f^2M^2 - e^2n^2 + (fM - en - e)^2 + f^2n^2 + \rho(2f - \rho) + 2f\,dn + e(2dn - e) = M^2\kappa$. For $\delta = 1$ ($t = n - 1$, $r = n + 1$, $q = n$, $\rho = 0$) the same identity reads $f^2M^2 - e^2(n-1)^2 + (fM - e(n+1))^2 + f^2(n-1)^2 + (4f^2 - f^2) + 4f(dn - f) + (dn + e)^2 - (dn - e)^2 = M^2\kappa$ with $eM = fn$ (checked by computer algebra). The canonical triple of Proposition 4.1 satisfies (K) under (H1), (H2). This proves the $W$ half of Theorem 3.1. The $T$ half is Section 3.3. □

*Remark* (where the hypotheses enter). (H1) gives $n \geq 1$; (H2) enters only through $dn \geq e + 1$, that is the single inequality $er < fq$ with $q = n$, $r = n + 1$: the apex $X_4$ of the corner removed from $\pi_5$ lies on $PO$ at or beyond $C'$. Everything else in (K) holds for all $M \geq 1$ with $dn \geq 1$ (for $\delta = 1$ the inequality $eq < ft$ needs $dn > f$, which follows from $de \geq f + 1$ since $n = em'$).

*Remark.* A shorter alternative to Step 1 (a winding-number chain lemma, audited inside the project, not in Lean) is given in Appendix A.2; it is not needed here.

## 5. Worked rows for $W_3(2, 5)$

**Example 5.1** ($W_3(2,5)$, 414 tiles)**.** $(e, f, M) = (2, 5, 3)$: $a = 10$, $b = 21$, $c = 25$, $\kappa = 46$, $d = 3$; $W_3(2,5)$ has sides $Mfb = 315$, $Me\kappa = 276$, $Mf^3 = 375$. $n = \lfloor 6/5 \rfloor = 1$, $\rho = 1$, $\delta = 0$, $(t, r, q) = (1, 2, 1)$. (K): $ft = 5 < eM = 6 < fr = 10$, $er = 4 < fq = 5$, $eq = 2 < ft = 5$. Scales: $v = 11$, $s_1 = 5$, $s_2 = 4$, $x = 3$, $u_1 = 3$, $u_2 = 1$. Points (Table 3), in $(\xi, y)$: $P = (0, 0)$, $O = (315, 0)$, $B = (0, 126)$, $A = (0, -150)$, $P_2 = (42, 0)$, $P_3 = (0, -20)$, $Q = (0, 105)$, $R = (0, 210)$, $C' = (84, 0)$, $B' = (40, 110)$, $V_2 = (50, 85)$, $X_4 = (105, 0)$, $U_1 = (30, 30)$, $U_2 = (80, 10)$; in $(u, w)$: $O = (7.5, 3)$, $B = (1.2, 3)$, $P_2 = (1, 0.4)$, $Q = (1, 2.5)$, $C' = (2, 0.8)$, $B' = (2, 3)$, $V_2 = (2, 2.5)$, $U_1 = (1, 1)$, $U_2 = (2, 1)$.

| | scales | formula | tiles | | scales | formula | tiles |
|---|---|---|---|---|---|---|---|
| $\pi_0$ | 15 minus 2 | $225 - 4$ | 221 | $\pi_3$ | 5 minus 4 | $25 - 16$ | 9 |
| $\pi_1$ | 11 | $11^2$ | 121 | $\pi_4$ | $as_1 \times cx$ | $2 \cdot 5 \cdot 3$ | 30 |
| $\pi_2$ | 5 | $5^2$ | 25 | $\pi_5$ | 3 minus 1 | $9 - 1$ | 8 |

Total $221 + 121 + 25 + 9 + 30 + 8 = 414 = 9 \cdot 46 = M^2(2f^2 - e^2)$. The tiling is drawn in Figure 4 (numbers in circles are the piece numbers); a list of the 414 tiles in exact coordinates is shipped and verified by an exact checker.

Three rows of the proof, by hand, for this example. *Separation* ($\pi_4|\pi_5$ by the line $w = q$). $w = \psi/(efb) = (2\xi + 5y)/210$. On $\pi_4$: $w(Q) = 525/210 = 2.5$, $w(U_1) = (60 + 150)/210 = 1$, $w(U_2) = (160 + 50)/210 = 1$, $w(V_2) = (100 + 425)/210 = 2.5$: all $\geq 1 = q$. On $\pi_5$: $w(U_1) = 1$, $w(P_2) = 84/210 = 0.4$, $w(C') = 168/210 = 0.8$, $w(U_2) = 1$: all $\leq 1$. So the line $w = 1$ separates the two (the pieces meet along $U_1U_2$). *Containment* ($\pi_5$, vertex $U_2$, in $\Omega = OAB$). The three sides: $\xi \geq 0$: $80 \geq 0$; $w \leq M$: $w(U_2) = 1 \leq 3$; $ef\xi - by \leq ef^2bM$: $10 \cdot 80 - 21 \cdot 10 = 590 \leq ef^2bM = 10 \cdot 5 \cdot 21 \cdot 3 = 3150$. Equivalently, as in Step 1: $U_2 \in [C', B']$, since on $u = 2$ its level $w = 1$ lies in $[0.8, 3]$. *Convexity and orientation* ($\pi_5 = U_1P_2C'U_2$). The edge vectors in $(\xi, y)$ are $U_1 \to P_2 = (12, -30)$, $P_2 \to C' = (42, 0)$, $C' \to U_2 = (-4, 10)$, $U_2 \to U_1 = (-50, 20)$, and the consecutive cross products $\xi y' - y\xi'$ are $12 \cdot 0 - (-30) \cdot 42 = 1260$, $42 \cdot 10 - 0 = 420$, $(-4)(20) - 10(-50) = 420$, $(-50)(-30) - 20 \cdot 12 = 1260$: all positive. (In the

proof convexity is automatic, since each piece is an intersection of half-planes.) *A shape row:* $P_2 - U_1 = (12, -30) = eu_1(e, -f) = 6\,(2, -5)$, of length $6f = 30 = a\,u_1$ since $au_1 = 10 \cdot 3$.

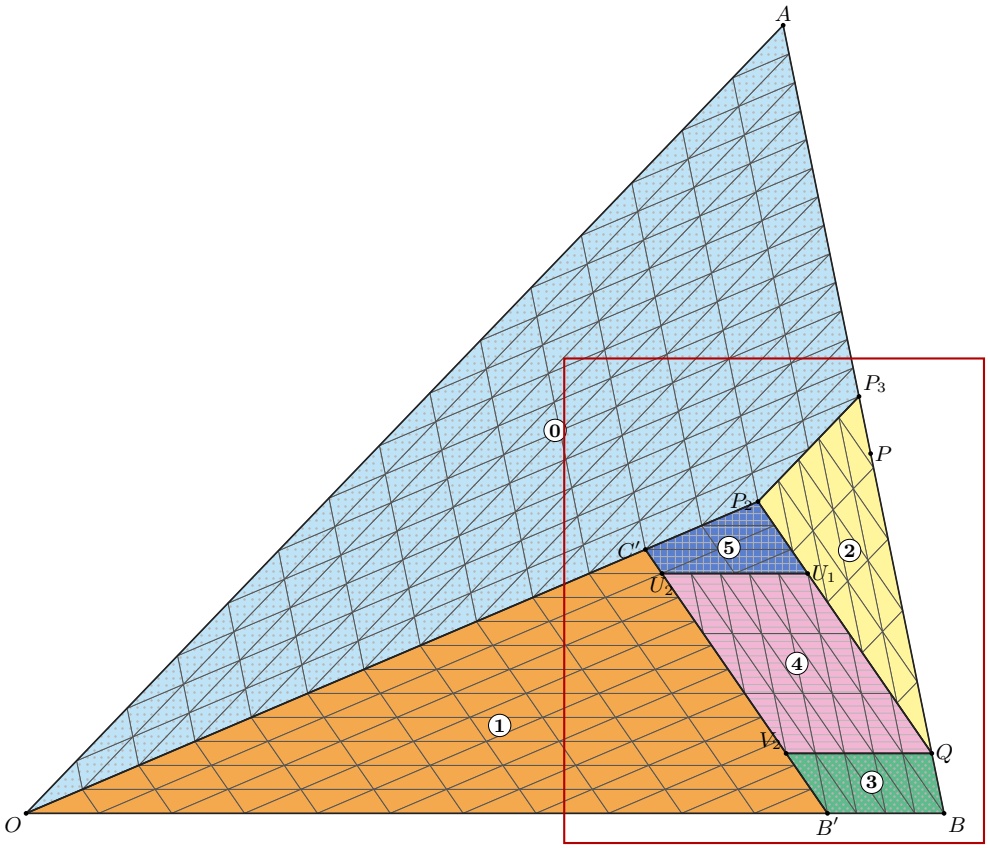
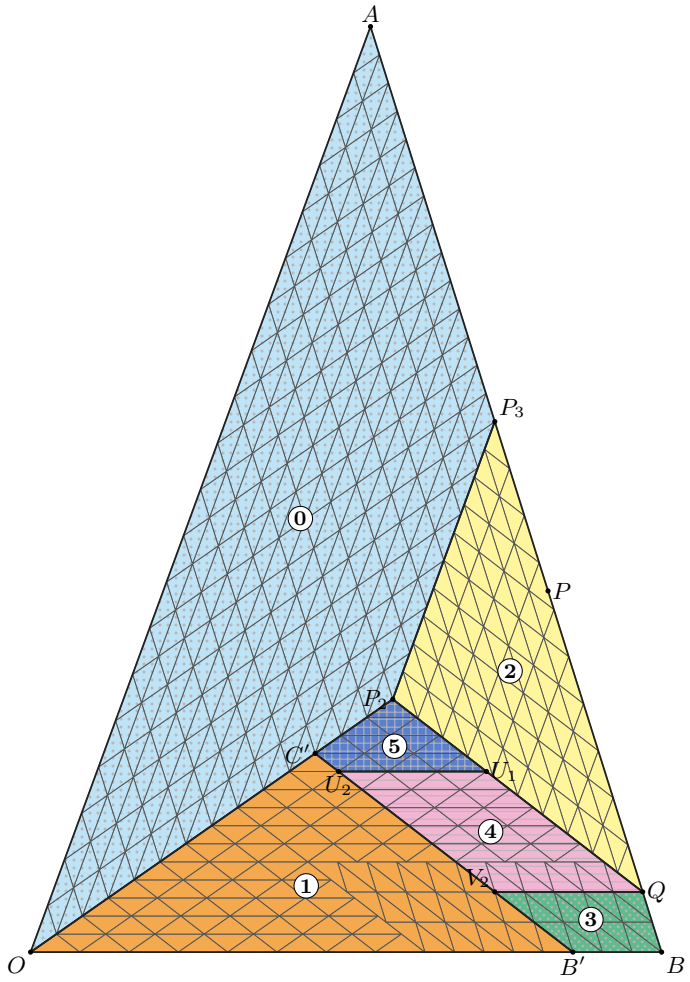



FIGURE 4. The layout for $W_3(2, 5)$ (tile $(10, 21, 25)$, 414 tiles, $(M, n, t, r, q) = (3, 1, 1, 2, 1)$). Colours are pieces, numbered 0–5 as in Table 4 (0 light blue, 1 orange, 2 yellow, 3 green, 4 pink, 5 dark blue); the thin lines are the 414 tiles of the exact tiling. The large pieces $\pi_0$ and $\pi_1$ carry 342 of the 414 tiles; the four small pieces lie in the red box (compare Figure 3).

**Example 5.2** ($W_4(3, 5)$, 656 tiles: below the old threshold, $M = 4 < 5 = f$). $(e, f, M) = (3, 5, 4)$: $a = 15$, $b = 16$, $c = 25$, $\kappa = 41$; $n = \lfloor 12/5 \rfloor = 2$, $\rho = 2$, $\delta = 0$, $(t, r, q) = (2, 3, 2)$; $v = 11$, $s_1 = 5$, $s_2 = 3$, $x = 4$, $u_1 = 4$, $u_2 = 1$; counts $364 + 121 + 100 + 16 + 40 + 15 = 656 = 16 \cdot 41$. See Figure 5.

**Example 5.3** (the case $\delta = 1$: $(e, f, M) = (2, 5, 5)$, $M = f$). Here $f \mid eM$: $n = 2$, $\rho = 0$, $\delta = 1$, so $(t, r, q) = (1, 3, 2)$ and $v = 19$, $s_1 = 10$, $s_2 = 5$, $x = 1$, $u_1 = 8$, $u_2 = 4$; counts $621 + 361 + 25 + 75 + 20 + 48 = 1150 = 25 \cdot 46$. (K): $5 < 10 < 15$, $6 < 10$, $4 < 5$. Here the strip is two tile-rows wide ($r - t = 2$).

**Example 5.4** (no layout: $(e, f, M) = (3, 5, 3)$). $d = 2$, $n = \lfloor 9/5 \rfloor = 1$, $dn = 2 \le 3 = e$, so by Proposition 4.1 there are no integers $t, r, q$ with (K): the canonical triple $(1, 2, 1)$ gives $u_2 = fq - er = 5 - 6 = -1 < 0$. This does not show that $W_3(3, 5)$ is untiled (it is an open case, Section 6).




**0** $\triangle OAP$ of scale $fM$ minus corner of scale $et$: $f^2M^2 - e^2t^2 = 364$

**1** triangle $OB'C'$ of scale $v = fM - er$: $v^2 = 121$

**2** triangle of scale $ft$: $f^2t^2 = 100$

**3** triangle of scale $s_1 = f(r-t)$ minus corner of scale $s_2 = fr - eM$: $s_1^2 - s_2^2 = 16$

**4** parallelogram $as_1 \times cx$, $x = ft - eq$: $2s_1x = 40$

**5** triangle of scale $u_1 = fq - et$ minus corner of scale $u_2 = fq - er$: $u_1^2 - u_2^2 = 15$

$t = n - \delta$, $\delta = [f \mid eM]$, $r = n + 1$, $q = n$, $n = \lfloor eM/f \rfloor$ here $(M, n, t, r, q) = (4, 2, 2, 3, 2)$ total $= M^2(2f^2 - e^2) = 656$

FIGURE 5. The same layout for $W_4(3, 5)$ (656 tiles of $(15, 16, 25)$, $(M, n, t, r, q) = (4, 2, 2, 3, 2)$). Colours and numbers as in Figure 4; counts $364 + 121 + 100 + 16 + 40 + 15$.

## 6. Consequences, and what stays open

6.1. **Effect on the census of tile counts.** Let $F$ be the set of those numbers $M^2(2f^2 - e^2)$ and $M^2(3f^2 - e^2)$, for the triples $(e, f, M)$ of Theorem 3.1, that are non-squarefree and $\le 10^6$. By the theorem all of them are tile counts. The census of the companion covers the non-squarefree integers $N$ with $2 \le N \le 10^6$ (the squarefree ones are treated separately there and are not part of this count). Among them the set $F$ has 44 495 elements (28 946 of the form $W$, 18 431 of the form $T$, with 2 882 values of both forms; largest $f$ is 251), none of them is excluded by the

exclusion theorem of the earlier release, 18 357 were known tile counts by other constructions, and 26 138 **moved from "undecided" (neither known to be a tile count nor excluded by the exclusion theorem of the earlier release) to "tile count"** (each such value is an instance of the Lean theorem `thmW`, which is universal in $(e, f, M)$; the count 26 138 itself comes from a program, [COMPUTED, EXACT], and the "undecided" status comes from the companion census). The census itself, the other constructions and the conjecture that organises them are in the companion paper [6]. Two examples: $414 = 2 \cdot 3^2 \cdot 23$ and $639 = 3^2 \cdot 71$ were undecided before ($414 = W_3(2, 5)$, $639 = T_3(2, 5)$).

6.2. **What stays open.**

- $e < f/2$**: nothing open for** $W_M$ **(given the import (I1), Corollary 3.2);** for $T_M$ the scales with $2Me \geq f$ and $M < M^*(e, f)$ are neither proved tiled nor excluded (for example $T_3(2, 11)$).
- $e > f/2$**, the "no" side.** The exclusion imported here is only (I2), $M(f - e) \leq e$. So for each row $(e, f)$ with $e > f/2$ the scales $M$ neither covered by Theorem 3.1 nor excluded by (I2) are exactly those with $e/d < M < f/d + 1$, $d = f - e$ (for $e > f/2$ the condition $Me \geq f$ holds for all $M \geq 2$); the interval has length exactly 2, so it contains at most two integers: at most two undecided scales per row. (That the interval holds at most two integers is proved: it is an open interval of length 2, which contains at most two integers.) [CONJECTURE] *Conjecture* $W^*$*:* for coprime $e, f$ with $2 \leq e \leq f - 2$, $W_M(e, f)$ is tiled iff $M \geq M^*(e, f)$ (for $e > f/2$: iff $(M - 1)(f - e) \geq f$). Examples, relative to the imports (I1)–(I3) only: $W_2(3, 5)$, $W_3(3, 5)$, $W_2(5, 8)$. For $W_2(5, 8)$ ($412 = 4 \cdot 103$ tiles) a restricted exact computer search found no tiling (evidence only, not formal, [COMPUTED, EXACT]).
- $e = 1$ **and** $e = f - 1$ lie outside Theorem 3.1; for $W_M$ the threshold is $M = f$ by (I3); for $T_M(1, f)$ the exclusion $2M \leq f$ of (I1) leaves, for example, $T_3(1, 4)$ open.
- **The** $T$ **half in general:** only the tile count and the glue; the converse of the glue is unknown.
- **Nothing is claimed about the prime case or any integer outside** $F$**.**

*Outside check.* The Lean statements and proofs were compared and audited inside this project only; no reader outside the project has checked them.

## 7. Credits and disclosure

*Credit.* The targets $W_M, T_M$, the $W/\beta$ framework and the tilings for $M \geq f$ are due to D. Paliy and to M. Beeson and G. Roşu (the tile with $3\alpha + 2\beta = \pi$ is Beeson's case III [1]); the isosceles-apex frame at the point $P$ with $OP = OB$ is the one of Beeson–Roşu; the six-piece layout was found by our own computer search and is proved here, and the lattice lines $\ell_k$ of Section 2.4 are our way of reading it. The companion paper [6] credits, statement by statement, B. Herdt, Y. X. Zhang, M. Laczkovich, J. P. Harries, and Snover–Waiveris–Williams, whose work enters there rather than here. *AI.* The work was done with the help of an AI system (Claude, Anthropic); the questions and directions were the author's, and each claim carries the label saying how it was checked (explicit tilings are checked by an exact program, not by Lean). Statements about Paliy's work refer to his public repository [3]; we have read his documents and checked some examples; nothing here claims priority over him for the results listed as his.

## Appendix A. Verification aid: what was computed, and the older proof

This appendix is not needed to follow Section 4.

A.1. **What is checked how.**

| statement | by what |
|---|---|
| Frame: coordinates of Table 3, the metric (1), the five directions | hand (Section 2.4); exact arithmetic against the older complex-number description (Appendix A.3) for all coprime $e, f$ with $f \le 24$, $M < 3f$ (all $f \le 40$ with sampled $M$), canonical and random $(t, r, q)$: no mismatch (`code/check_thmW_frame.py`, block A) |
| Side lengths of Table 5, parallelism, levels of the cut lines | hand; exact: 18 276 cases, no mismatch (block B) |
| Step 1 (six pieces: separation, containment, convexity, orientation, areas add up) | hand; exact: 201 886 triples $(e, f, M)$, $f \le 60$, $M < 5f$: no failure under (H1), (H2); none passes outside (block D) |
| (K) $\Leftrightarrow$ the 16 sign conditions of the older proof; layout exists iff $dn \ge e+1$; canonical triple works whenever any works | hand (Prop. 4.1); exact enumeration of all $(t, r, q)$ for $f \le 12$, $M < 3f$, 17.9 million triples (block E) |
| Lemma 3.5 and every step of its proof | hand; exact: all coprime $2 \le e \le f-2$, $f \le 300$, $M \le 3f$, 16.2 million triples (block G) |
| Count identity of Section 4.5, and the three factor identities $C_8 = fC_7 + eC_6$, $C_9 = fC_2 + C_7$, $C_6' = C_6 + eC_2$ | computer algebra (block H, sympy) |
| The Lean proof (the older route) | Lean (`ThmWIneq0`–`4`, `ThmWGeo0`–`3`, `ThmWParams`, `ThmWNat`, `ThmWFinal`); the cross-product tables of Appendix A.3 are produced by `code/check_thmW_algebra.py` |
| Packing: finitely many convex pieces, each tiled, disjoint interiors, area sum = area of target $\Rightarrow$ tiling | Lean (general theorem in the library) |
| Explicit tilings (e.g. 414, 656, 639, 801, 3950, 13 888, 57 232 tiles) | exact checker on exact coordinates: congruence, disjointness, area, hull a triangle ([COMPUTED, EXACT]; not Lean) |
| Not in this paper's Lean statement | the tiling of $T_M$ (glue (I5), Lean in the earlier release); the exclusions (I1)–(I3) |

Consistency with the proved exclusions: (H1), (H2) imply $M(f-e) > e$, $Me \ge f$ and $2Me \ge f$, so no instance of the theorem contradicts (I1) or (I2). Reproduction: `python3 code/check_thmW_frame.py ALL 60` (about one minute on one core) and `python3 code/check_thmW_algebra.py 60 -tex tables.tex`.

A.2. **Alternative to Step 1: the chain lemma, and certificates for (K).** *Hand-proved, audited inside the project (verdict: sound, with four repairs, all applied here); not in Lean.* Step 1 above is a direct case analysis. A shorter argument replaces containment, separation and the area identity by one lemma (winding-number additivity; the integer frame is due to a study inside this project).

**Lemma A.1** (chain lemma)**.** *Let $T$ and $\Pi_1, \dots, \Pi_k$ be triangles or quadrilaterals with counterclockwise vertex lists, all with* strictly *positive turns (cross products of consecutive edge vectors). (For polygons with five or more vertices positive turns do not imply convexity.) Then each $\Pi_i$ is a convex polygon: for a 3- or 4-gon whose exterior angles all lie in $(0, \pi)$ and sum to $2\pi$, the*

*boundary is a simple closed convex curve. If, as formal sums of oriented segments,* $\sum_i \partial\Pi_i = \partial T$*, where the equality means equal oriented multiplicity on every line (collinear segments add:* $[p,q]+[q,r]=[p,r]$ *in any order of the points, and* $[p,q]+[q,p]=0$*), then the* $\Pi_i$ *have pairwise disjoint interiors, lie in* $T$*, and cover* $T$*.*

*Proof.* Winding numbers are additive in the boundary chain; a convex counter-clockwise polygon has winding number 1 inside and 0 outside, so $\sum_i \mathbf{1}_{\Pi_i} = \mathbf{1}_T$ off finitely many lines. If two interiors met, the sum would be $\geq 2$ on an open set; if some $\Pi_i$ left $T$, it would be $\geq 1$ off $T$. □ (Caution: "each piece inside $T$ and the areas add up" does *not* imply disjointness; and the vanishing of a sum of edge *vectors* is not chain equality.) *The chain identity* holds for all integers $e, f, M, t, r, q$, because the points of Table 3 lie on five lines: $AB$ ($\xi = 0$: $A, P_3, Q, B$), $OP$ ($y = 0$: $O, P_2, C'$), $OB$ ($e\xi + fy = efbM$: $O, B, B'$), $\ell_t$ ($\varphi = efbt$: $P_2, Q, U_1$), $\ell_r$ ($\varphi = efbr$: $C', B', U_2, V_2$); in each the parameters cancel (for $U_1$: $f \cdot ef(ft-eq) + e \cdot ef(fq-et) = efbt$). Then on each line the oriented edges cancel: $C' \to O+O \to P_2+P_2 \to C' = 0$; $B \to Q+Q \to P_3+P_3 \to A = B \to A$; $O \to B'+B' \to B = O \to B$; $P_2 \to Q+Q \to U_1+U_1 \to P_2 = 0$; $V_2 \to B'+B' \to C'+C' \to U_2+U_2 \to V_2 = 0$; and $P_2P_3$, $QV_2$, $U_1U_2$ are shared by two pieces with opposite orientation. *What remains* is strict positivity of every turn: each is a positive monomial times a product of the factors $t$, $r-t$, $eM-ft$, $fr-eM$, $fM-er$, $ft-eq$, $fq-er$, $fM-et$, $fq-et$. Five are independent: $t$, $C_3 = eM-ft$, $C_4 = fr-eM$, $C_6 = fq-er$, $C_7 = ft-eq$; the others follow from $f(r-t) = C_3 + C_4$, $fq-et = C_6 + e(r-t)$, $fM-et > 0$ (as $et < eM < fM$) and the identity $e\,C_5 = f\,C_3 + f\,C_7 + e\,C_6$ for $C_5 = fM-er$ (and $C_5 = u+1+fs$ below, positive anyway). *Certificates.* Put $\mu := dn-e-1$ ($\geq 0$ by Lemma 3.5), $\rho' = f-1-\rho \geq 0$, $M = n+s$, so $\rho = es-dn$. For the canonical triple: $C_3 = \rho + f\delta$, $C_4 = \rho'+1$, $C_5 = \mu+1+fs$, $C_6 = \mu+1$, and $C_7 = \mu+1+e$ if $\delta = 0$; if $\delta = 1$ ($M = fm'$, $n = em'$, $e-1 = 1+x$, $d-1 = 1+y$, $x, y \geq 0$, $x+y \geq 1$) then $C_7 = (m'-1)ed + x + y + xy \geq 1$. Each is a sum of non-negative terms with a positive constant. Checked exactly for all 982 658 admissible triples ($f \leq 120$, $M \leq 3f$); the 27 322 non-admissible triples in that range each violate a turn, so for these triples the canonical layout fails (compare Proposition 4.1, which is the actual sharpness statement). The Lean proof of this paper does not use the chain lemma: it proves the placement conditions the older way (the sixteen named factors of Appendix A.3, 143 generated sign lemmas, kernel-checked). A Lean version of the chain lemma is planned, not done. Still needed for Step 2, unchanged: that each piece is a union of tiles (the edge-length identities of Table 5 and Proposition 2.1).

A.3. **The older proof: complex coordinates and cross-product tables.** The first version of this paper, and the Lean proof, use complex numbers with $O = 0$ and $OB$ the positive real axis. With $D = 4f^2 - e^2$ write $\xi + i\sqrt{D}\,\zeta$ as $[\xi, \zeta]$ and put

$$\varepsilon = e^{i\alpha} = \left[\tfrac{\kappa}{2f^2}, \tfrac{e}{2f^2}\right], \quad \eta = e^{i\beta} = \left[\tfrac{e(3f^2-e^2)}{2f^3}, \tfrac{b}{2f^3}\right], \quad \omega = \overline{\varepsilon\eta} = e^{-i(\alpha+\beta)},$$

with $\varepsilon^3\eta^2 = -1$, $a + b\omega = c\bar\eta$, $f\omega + e\varepsilon = f\bar\eta$ (each checked by expansion). Then $P = fbM\varepsilon$ and the frame is $z = P - \xi\,\varepsilon + y\,\omega$: the first axis points from $P$ to $O$ (direction $-\varepsilon$) and the second from $P$ to $B$ (direction $\omega$). In these terms the points of Table 3 are $B = fbM$, $A = f^3M\varepsilon^2$, $P_2 = b(fM-et)\varepsilon$, $P_3 = P - eat\,\omega$, $Q = P + fbt\,\omega$, $R = P + fbr\,\omega$, $C' = bv\,\varepsilon$, $B' = cv$, $V_2 = P + f^3t\,\bar\eta - ef^2r$, $X_4 = P - fbq\,\varepsilon$, $U_1 = P - ef^2t + ef^2q\,\bar\eta$, $U_2 = P - ef^2r + ef^2q\,\bar\eta$ (exact check against the frame: block A). The cross product $\langle u, v\rangle = \frac{1}{\sqrt{D}}\mathrm{Im}(\bar u v)$ is $1/\sqrt{D}$ times the usual one. The Lean proof shows that each cross product needed for the containment of a vertex in $OAB$, for the separation of two pieces and for the convexity of a piece equals a positive

constant times a product of the *named factors*

$$C_2 = r - t,\ C_3 = eM - ft,\ C_3' = fM - et,\ C_4 = fr - eM,\ C_5 = fM - er,$$
$$C_6 = fq - er,\ C_6' = fq - et,\ C_7 = ft - eq,\ C_8 = f^2t - e^2r,\ C_9 = fr - eq,$$
$$C_{10} = M - q,\ E_1 = bM + eft,\ E_2 = eMf + bt,\ E_3 = br + e^2t,$$
$$L_A = bfM - e\kappa t + f^3q,\ L_B = bfM - e\kappa r + f^3q,\ L_C = efbM - e^2\kappa r + f^4t,$$

and that these are positive for the canonical triple. Under (K) they all follow from the four inequalities: $C_3, C_4, C_6, C_7 > 0$ are (K) itself; $C_2 > 0$, $C_{10} > 0$ and $C_3', C_5 > 0$ are (4); $C_6' = C_6 + eC_2$, $C_8 = fC_7 + eC_6$, $C_9 = fC_2 + C_7$ are sums of positive terms; $E_1, E_2, E_3$ are sums of products of positive numbers; $L_A, L_B, L_C$ express that $U_1$, $U_2$, $V_2$ lie on the segments $[P_2, Q]$, $[C', B']$ of Step 1, which is automatic (verified exactly: block E). The three tables (separation, containment, convexity; about 100 rows) are generated by `python3 code/check_thmW_algebra.py 60 -tex tables.tex` and are not reproduced here (they are in the earlier draft in the repository history); each row is a bilinear expansion with the sine table $\langle 1, \varepsilon\rangle = \frac{e}{2f^2}$, $\langle 1, \omega\rangle = -\frac{1}{2f}$, $\langle 1, \bar\eta\rangle = -\frac{b}{2f^3}$, $\langle \varepsilon, \omega\rangle = \langle \varepsilon, \bar\eta\rangle = -\frac{1}{2f}$, $\langle \omega, \bar\eta\rangle = \frac{e}{2f^2}$. The values of the sixteen factors of the older proof for the three examples (all positive):

| $(e,f,M)$ | $C_2$ | $C_3$ | $C_3'$ | $C_4$ | $C_5$ | $C_6$ | $C_6'$ | $C_7$ | $C_8$ | $C_9$ | $C_{10}$ | $L_A$ | $L_B$ | $L_C$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $(2,5,3)$ | 1 | 1 | 13 | 4 | 11 | 1 | 3 | 3 | 17 | 8 | 2 | 348 | 256 | 887 |
| $(3,5,4)$ | 1 | 2 | 14 | 3 | 11 | 1 | 4 | 4 | 23 | 9 | 2 | 324 | 201 | 1103 |
| $(2,5,5)$ | 2 | 5 | 23 | 5 | 19 | 4 | 8 | 1 | 13 | 11 | 3 | 683 | 499 | 1123 |

## Appendix B. Data, Lean index, reproduction

B.1. **The thirteen witnesses with $M < f$ found by the first search.** Found first by a computer search, now instances of Theorem 3.1; each is a Lean theorem (module `Draft/DebtBWTilable`) with the closure upward in $M$ (I5). Columns: tiles of $W_M$ and of $T_M$.

| $(e,f,M)$ | $W_M$ | $T_M$ | $(e,f,M)$ | $W_M$ | $T_M$ |
|---|---|---|---|---|---|
| $(3,5,4)$ | 656 | 1056 | $(7,12,4)$ | 3824 | 6128 |
| $(4,7,4)$ | 1312 | 2096 | $(7,10,5)$ | 3775 | 6275 |
| $(5,8,4)$ | 1648 | 2672 | $(7,9,6)$ | 4068 | 6984 |
| $(5,7,5)$ | 1825 | 3050 | $(8,11,5)$ | 4450 | 7475 |
| $(5,11,3)$ | 1953 | 3042 | $(9,11,7)$ | 7889 | 13 818 |
| $(5,9,4)$ | 2192 | 3488 | $(6,11,4)$ | 3296 | 5232 |
| $(7,11,4)$ | 3088 | 5024 | | | |

B.2. **Shipped tilings.** Directory `data/witnesses/v3_exact/`; files are named by tile count, kind and parameters (for example `tiling_414_W_2_5_M3`, `tiling_656_W_3_5_M4`, `tiling_639_T_2_5_M3`, `tiling_1056_T_3_5_M4`, each with the suffix `.exact.txt.gz`). Each begins with a comment line (tile count, tile sides); each further line gives the three vertices of one tile in exact coordinates $(X, Y\sqrt{D})$. Kinds `W` and `T`: 37 and 15 files.

B.3. **Lean index.** All names in namespace `Erdos634.`; aggregate `AllThmW`. Theorem 3.1: `Draft.ThmW.thmW` (modules `ThmWLib`, `Lib2`, `Hyp`, `Sc`, `Pts`, `Ineq0`–`4`, `Geo0`–`3`, `Core`, `Nat`, `Params`, `Final`); instances `inst_414`, `inst_656`, `inst_639`, `inst_801`, `inst_3088`, `inst_27324`. Counts: `Draft.DebtB.counts_of_wTilable`; glue: `CollarG.betaTilable_of_wTilable`; 13 witnesses: `Draft.DebtB.closure_656` etc.; ladder: `Subdivision.ladderDissection`.

B.4. **Reproduction.**

```
python3 code/check_thmW_frame.py ALL 60                 # frame, layout, equivalences
python3 code/check_thmW_algebra.py 60 --tex tables.tex  # older algebra + tables
python3 code/lakeserial.py lean build Erdos634.AllThmW  # one module at a time
BTE=rust/bonfioli_tiling_engine/target/release/bte
$BTE verify data/witnesses/v3_exact/tiling_414_W_2_5_M3.exact.txt.gz
```

## References


[1] M. Beeson, *Triangle tiling III: the case* $3\alpha + 2\beta = \pi$, arXiv:1206.2229.

[2] M. Beeson and G. Roşu, *Tiling a triangle into a prime number of congruent triangles*, preprint, version of 16 October 2026.

[3] D. Paliy, repository `github.com/DenisUkranian/erdos-634-tilings`: the $W/\beta$ framework, caps and collars, tilings for $M \geq f$ (September–October 2026); unrefereed.

[4] V. Bonfioli, *Congruent dissections of a triangle: prime counts, the counts up to* 200*, and a family of isosceles targets*, release v1.13.1, Zenodo, October 2026 (concept DOI doi:10.5281/zenodo.23400433).

[5] V. Bonfioli, *Congruent dissections of a triangle: runs, windows and the corner*, version 2.2, release v1.16, Zenodo (same concept DOI).

[6] V. Bonfioli, *Conjecture C, the collar family and the census of tile counts, with computer evidence* (companion to Paper W), version 3.0, release v3.0.